

Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Size the fuel market, then test what Vopak can serve

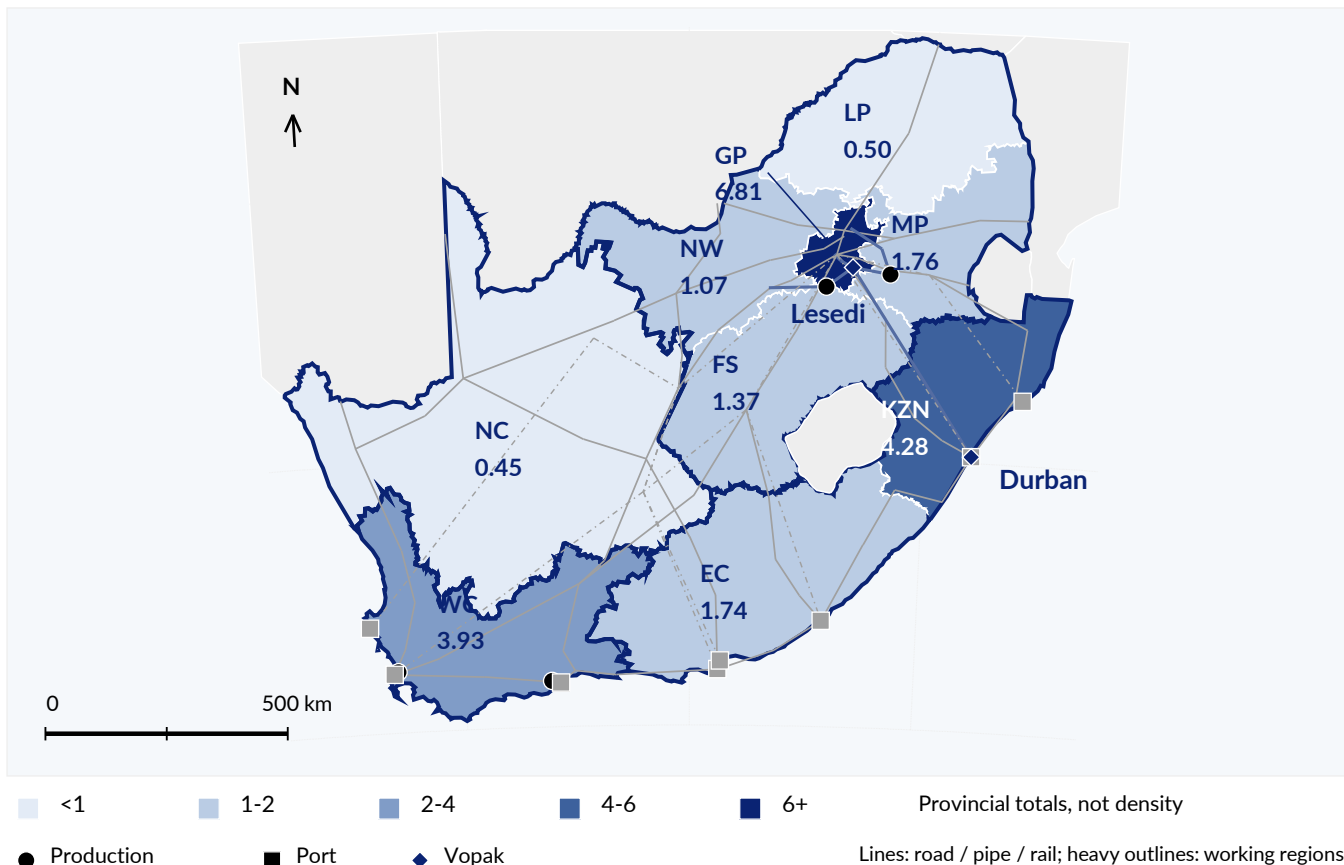
Situation → Complication → Resolution | evidence, actions and delivery roadmap

SCR reference	Evidence and required next step	Proposed lead	Timing and output	Deep dives
S1 Situation	Demand baseline and regional market Provincial sales locate demand. Refresh the complete year and reconcile historical totals.	Manish Nigel review	W1-W2 Sourced demand baseline	3 / 4
S2 Situation	Domestic supply and finished imports 2024 trade is sourced, but diesel imports conflict. Match production/stocks and identify import entry ports.	Manish Nigel access	W1-W3 Fuel balance + port bridge	5 / 6
C1 Complication	Demand drivers Vehicles, sectors, power and electrification move differently. Source the evidence first, then quantify fuel effects.	Manish Nigel review	W1 define; W2-W4 test Lever-to-demand bridge	7
C2 Complication	Refinery availability Capacity has fallen. Separate operating output from conditional redevelopment and agree plant scenarios.	Manish Henry / Nigel	W1-W3 Plant output + imports	8
C3 Complication	Competing supply routes Compare Durban, Matola and Walvis Bay for one customer. Test full route cost, capacity and customer access.	Manish Nigel / Henry	W1 scope; W2-W4 test Delivered R/litre + access	9 / 10
R1 Resolution	Vopak share and additional demand Match unique deliveries to regional demand. Test additional volumes against physical and commercial access.	Nigel client data Manish analysis	W1 request; W2-W4 test Served + contestable litres	11 / 12 / 13
R2 Resolution	Storage and optional investment Use published tanks as context. Assess additional services only after the incremental customer-flow case.	Nigel priority Manish / Henry	W4-W6 if prioritised Flow-to-stock / investment	14 / 15

Petrol + diesel; jet separate. Owners and weeks are proposed. Open gaps and illustrations remain explicitly labelled.

Three provinces account for 69% of reported fuel sales

Petrol + diesel sales by province | 2022, bn litres/year



Sales totals, not density. Vopak locations highlighted; detailed operator inventory on pages 12 and 14. Jet excluded.

Source: DMPR 2022 sales; geoBoundaries/OCHA/MDB 2020 provinces (CC BY 3.0 IGO); operator sources on slide 12.

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What the evidence shows

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Four reporting regions

Eastern coast: EC/KZN. Inland: GP/FS/LP/MP/NW. Western coast: WC. Other: NC. Boundaries are reporting groups.

03 | Sales locate the historical market

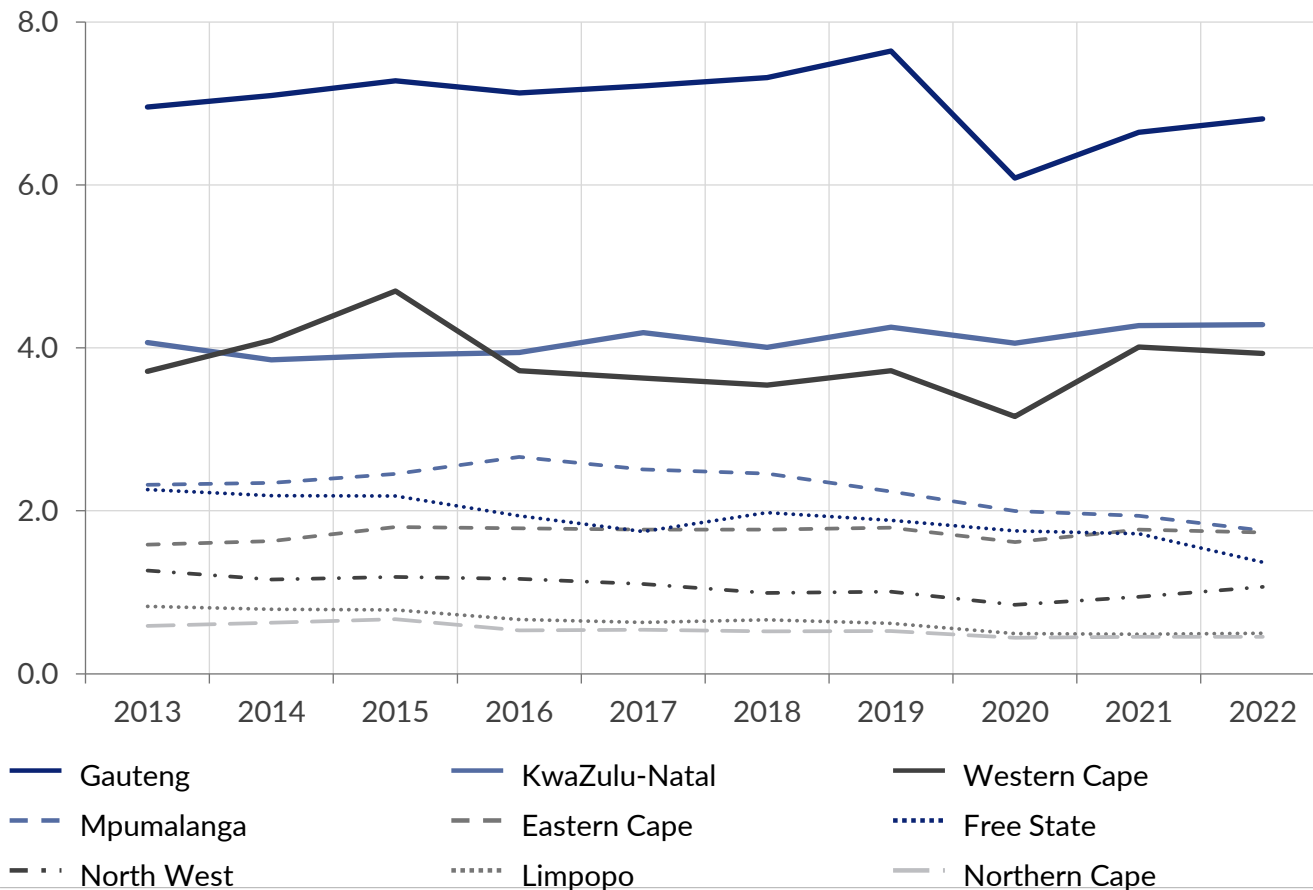
Colours show provincial sales totals, not density or customer catchments. Only Durban and Lesedi are labelled; competitor detail is on pages 12 and 14.

To resolve next

Manish: refresh a complete year and reconcile the source flags. Nigel: confirm the customer markets to test against Durban and Lesedi.

Provincial sales show different demand patterns over time

Annual provincial sales | 2013–2022, bn litres/year



What the evidence shows

01 | Gauteng remains the largest market

The reported series places Gauteng above the other provinces throughout 2013–2022. Volumes differ across the smaller markets.

02 | The series stops at 2022

2022 is the latest complete year in this extract. Only Q1 is available for 2023, so it is excluded from annual comparisons.

03 | Historical reconciliation is open

Provincial totals differ from national sales in six years: 2013, 2014, 2015, 2017, 2018 and 2021.

To resolve next

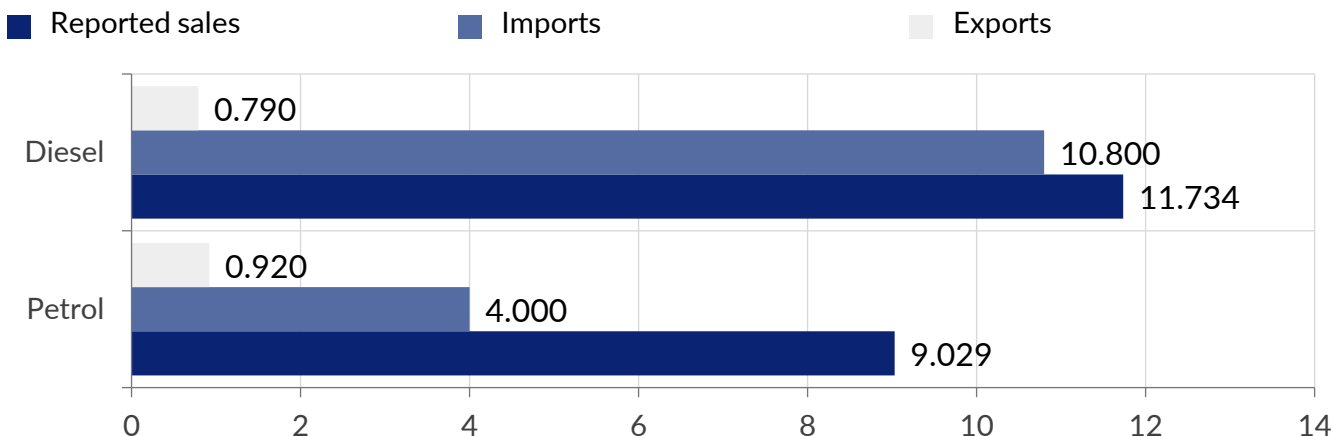
Manish: reconcile flagged workbook cells and source newer observed sales. Record revisions and incomplete periods before updating the outlook.

Imports are material, but the matched fuel balance is still open

National petrol/diesel | 2024, billion litres

Production + imports – exports – stock build = consumption

Compare consumption with reported sales; explain coverage and the residual.



Net imports: petrol 3.080, diesel 10.010; combined 13.090 bn litres.

Source flag | diesel imports

FIASA 14.793; government 10.800.
Difference -3.993 bn L. Preserve both;
investigate scope and vintage.

2024 accounting remains open

Sales less net imports = 7.673 bn L.
Balancing requirement, not production.
Production, stocks and coverage unresolved.

What the evidence shows

01 | Sales measure the domestic market

Reported petrol/diesel sales are a demand proxy. They are not imports plus exports; exports leave the national market.

02 | The balance needs matched inputs

Domestic production plus imports, less exports and stock build, gives consumption. Compare that result with reported sales and explain the residual.

03 | Diesel trade remains disputed

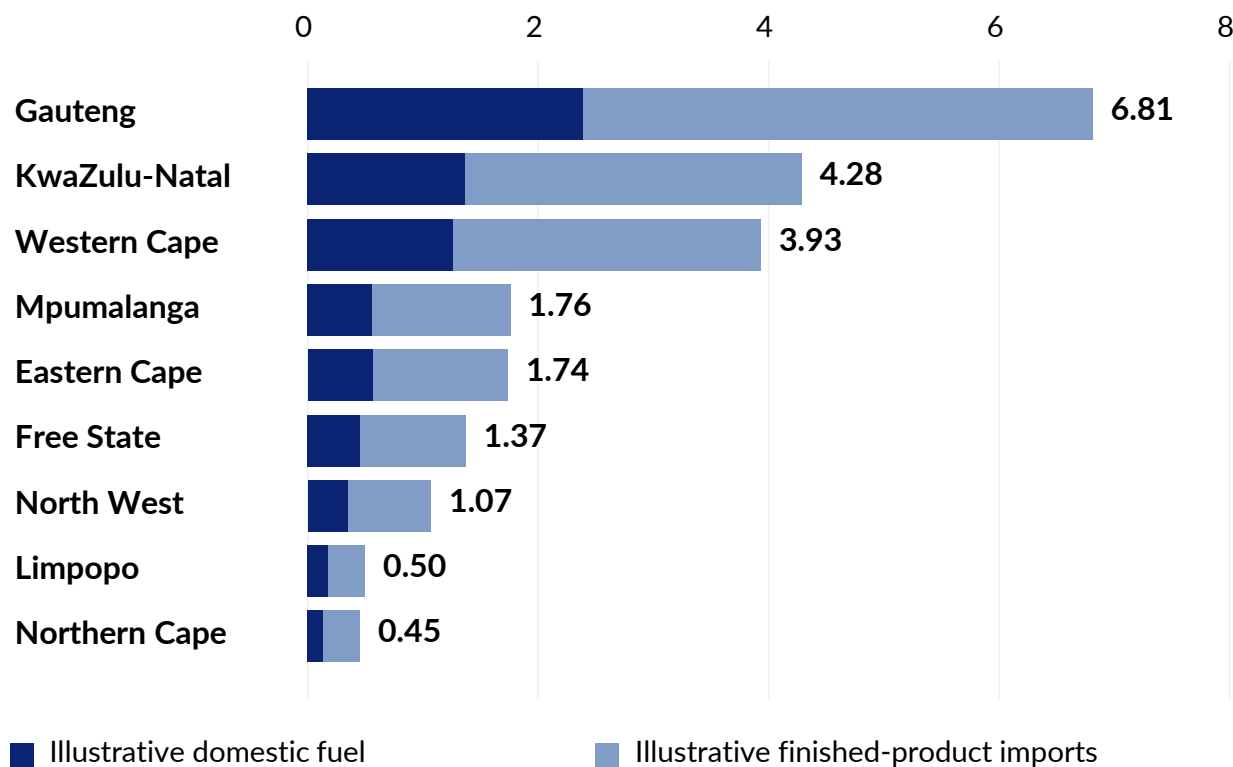
Government diesel imports are 10.8 bn litres; FIASA gives 14.793. Production and stock changes for 2024 remain open.

To resolve next

Manish: match trade, actual production and stock changes by product and year. Nigel: obtain product-specific port records.

Provincial demand is known; its supply origin still needs evidence

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

What the evidence shows

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

To resolve next

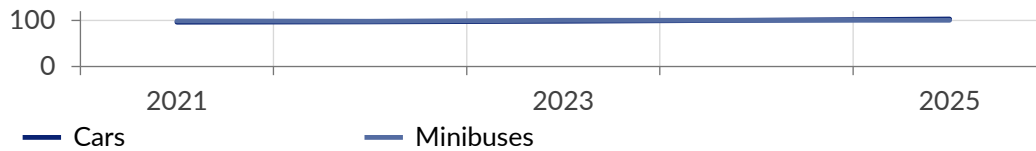
Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

Demand drivers move differently across vehicles and sectors

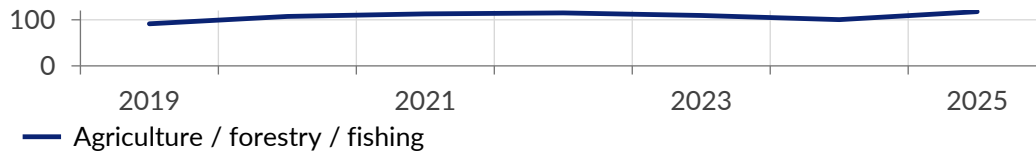
Potential demand levers | 2024 = 100; price panel uses nominal R/litre

Observed proxies, not fuel-volume effects. Activity axes 0–120; EV 0–400; prices 0–30 R/L.

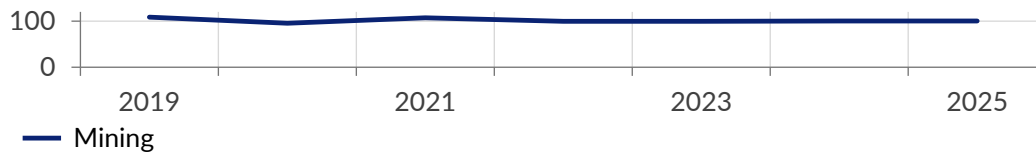
Passenger travel | all-fuel stock, Dec



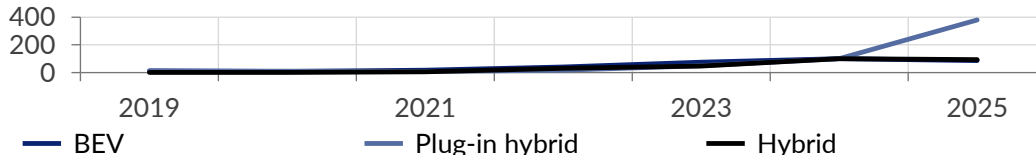
Agriculture | real value added, CY



Mining | real value added, CY



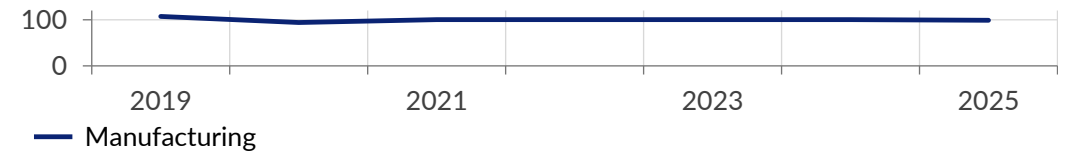
Electrification | annual new sales, CY



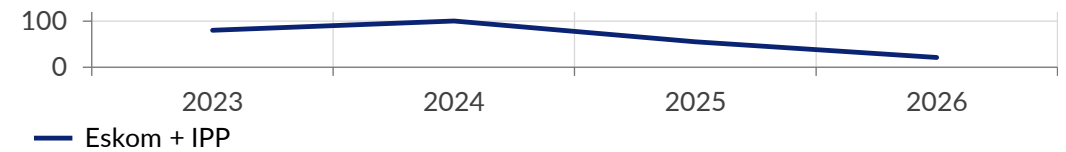
Freight demand | payload, CY



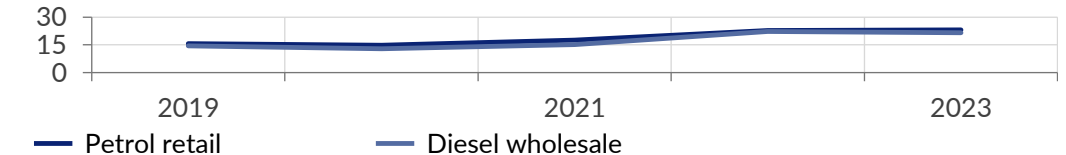
Manufacturing | real value added, CY



Diesel power | OCGT generation, FY



Price response | annual nominal R/litre



To quantify: mileage × fleet/fuel mix × litres/km; sector fuel intensity; OCGT litres/kWh; price elasticity. FY ends March. Price 2024 incomplete.

Source: NaTIS; Stats SA P0441 / P7162; Eskom; naamsa; Department fuel-price history. Scope and observations in notes.

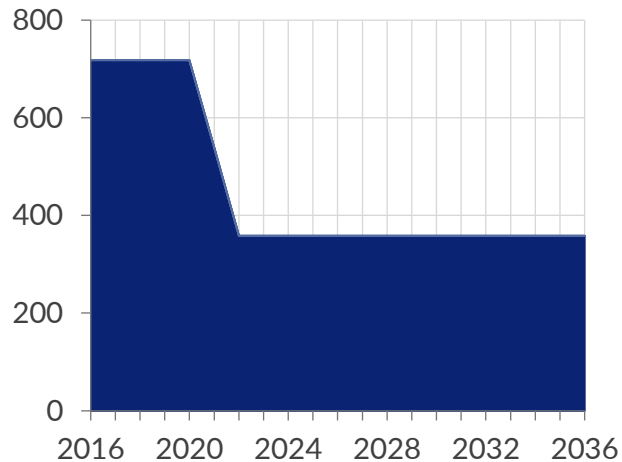
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Refinery capacity has fallen; redevelopment remains conditional

Published footprint + illustrative scenarios | thousand bbl/day

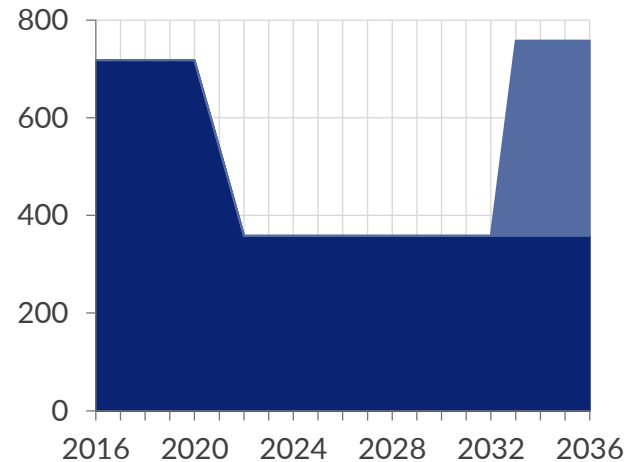
History 2016–2025 is sourced; 2026 onward is illustrative, not a forecast

A | No new refinery capacity



2036: 358 thousand bbl/day

B | Conditional CEF redevelopment



2036: 758 thousand bbl/day

B adds 400 from 2033; darker navy = existing footprint; lighter blue = conditional addition.

B assumes illustrative FID 2029 + 48 months; no sanctioned date. Both hold existing capacity flat.

What the evidence shows

01 | History shows a structural decline

Published total falls from 718 in 2016–2020 to 538 in 2021 and 358 from 2022–2025. Capacity does not measure realised output.

02 | Redevelopment is conditional

CEF proposes approximately 400 thousand bbl/day after FID. Construction is approximately 48 months; the 2029 FID here is authored for comparison.

03 | Product output still needs modelling

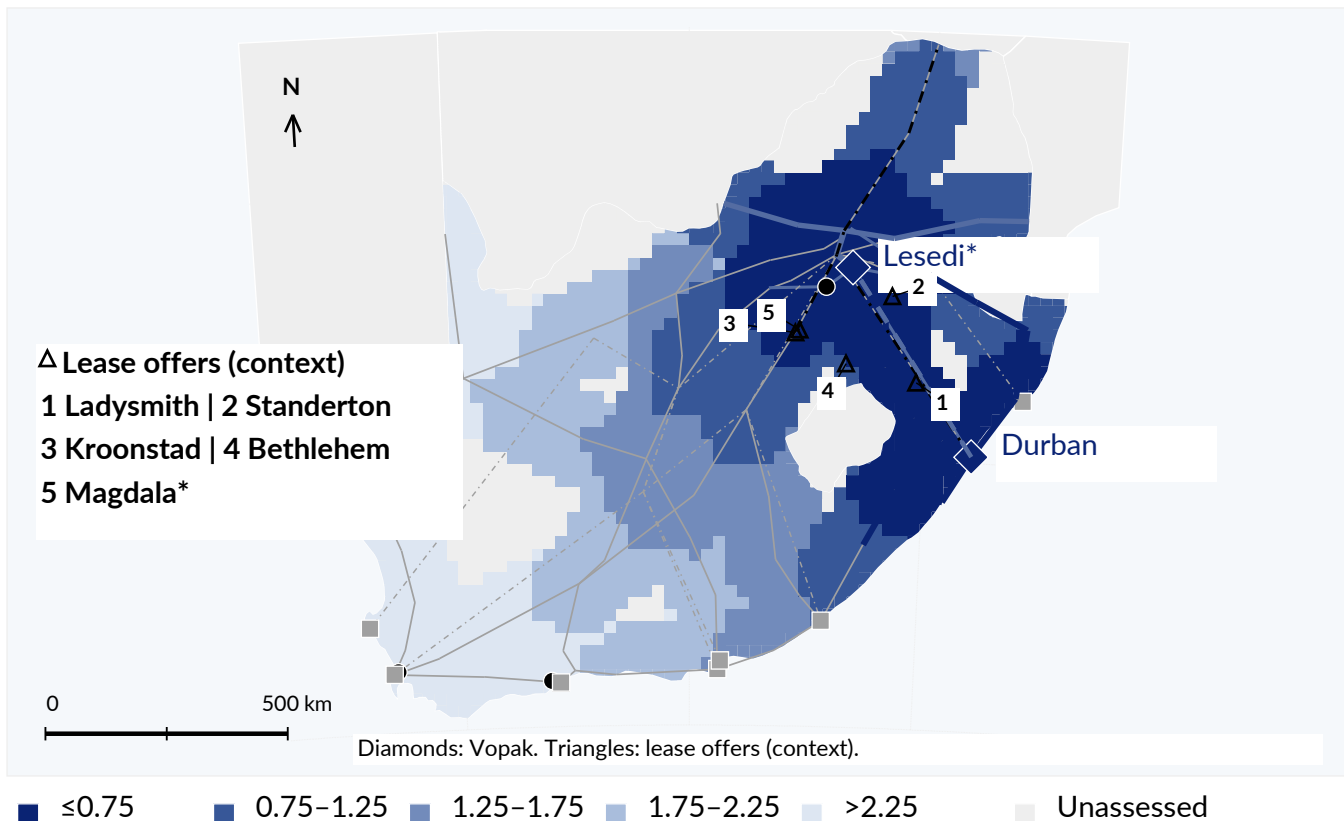
Source utilisation, downtime and product yields before translating capacity into domestic supply. Secunda/Natref cases remain open.

To resolve next

Nigel: agree scenario timing. Manish: reconcile plant capacity and output/yields. Henry: review technical feasibility and the Sasol gas-to-liquids mechanism.

Transport cost shapes the market Durban and Lesedi can reach

Illustrative road cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

Illustrative road cost only. Pipeline/rail access and commercial rights are not priced.

Source: Existing appendix routes; illustrative candidate connections, not verified catchments. Natural Earth 1:50m.

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What the evidence shows

01 | Darker means lower transport cost

Cells show example road cost from Durban or Lesedi. The rate is illustrative and excludes fuel price, storage and pipeline tariffs.

02 | Grey means unassessed

Grey areas have no assessed road connection under this illustration. They are not confirmed to be inaccessible.

03 | A cost surface is a first screen

Pipeline and rail lines are context. Route capacity, product compatibility and customer rights still determine serviceable demand.

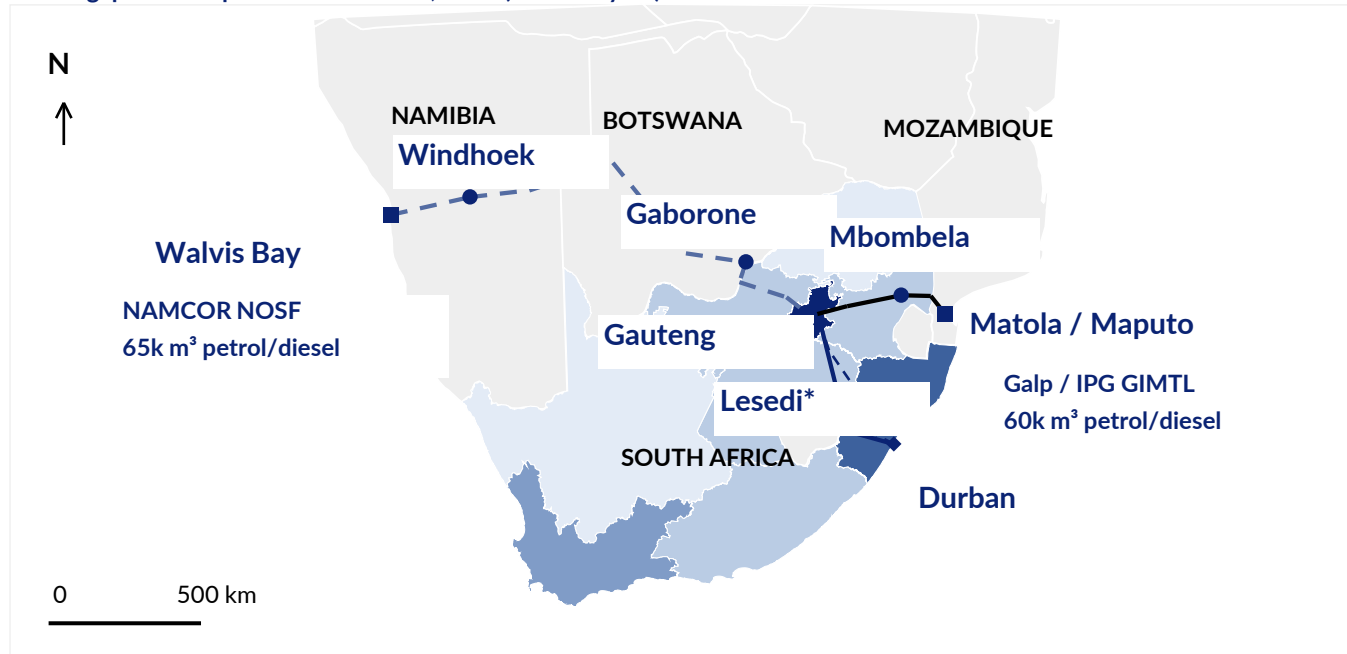
To resolve next

Nigel: select the first customer destination and cost threshold. Manish: replace example rates with dated route costs and access evidence.

Compare Durban, Matola and Walvis Bay for the same inland customer

Durban, Matola and Walvis Bay | schematic corridor comparison

Shading: provincial petrol + diesel sales, 2022 (bn litres/year)



What the evidence shows

01 | Vopak has a South African footprint

Durban and Lesedi are highlighted against 2022 provincial sales. Colour shows sales totals; it does not establish Vopak market dominance.

02 | Matola has published fuel storage

Galp/IPG GIMTL lists 60,000 m³ of petrol/diesel tanks, plus 6,000 m³ LPG excluded here. N4 is the eastern inland comparison route.

03 | Walvis has storage and handling

NAMCOR NOSF lists 75,000 m³ in total, including 65,000 m³ petrol/diesel. Trans-Kalahari is the western inland comparator.

To resolve next

Manish: verify available tanks, customer access and full delivered costs. Source Namibia/Mozambique demand before extending the heatmap; grey means unassessed.

Routes schematic; access unverified. Colour measures provincial totals, not density or Vopak share.

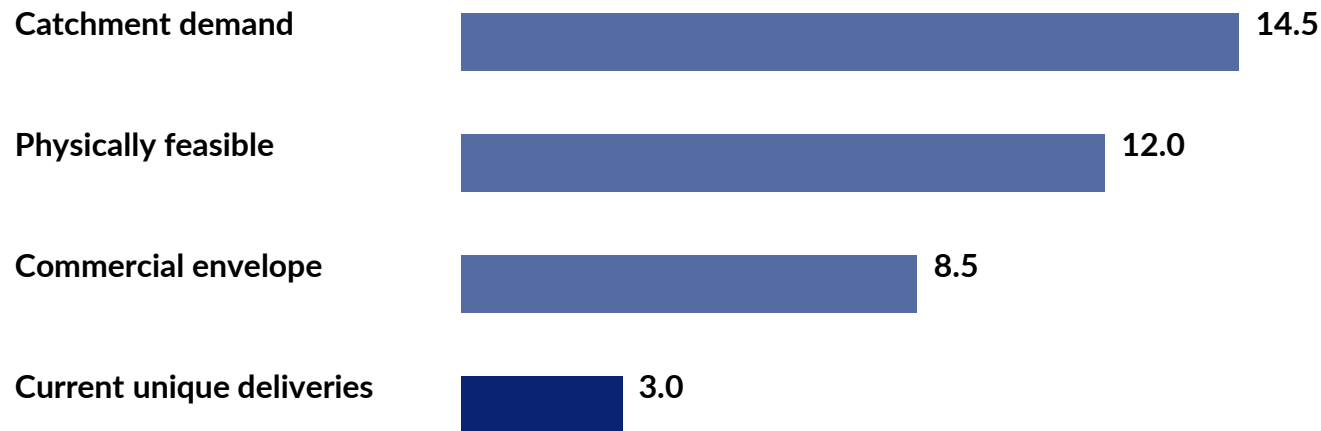
Source: DMPR 2022 sales; NAMCOR NOSF; Galp GIMTL; WBCG / TRAC corridors; Natural Earth. Capacities and URLs in notes.

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Separate market demand, accessible volume and unique Vopak deliveries

Market screens | illustrative bn litres/year

Successive screens; these volumes are illustrative



Count Durban–Lesedi transfers once

Durban 2.8 + Lesedi 2.0 – shared transfer 1.8 = unique 3.0

Demand ≠ terminal receipts ≠ storage capacity. Incremental opportunity is on page 13.

What the evidence shows

01 | Count customer demand once

Example receipts of 2.8 at Durban and 2.0 at Lesedi include a shared 1.8 transfer. Unique demand served is therefore 3.0 bn litres.

02 | Accessibility precedes opportunity sizing

Example catchment demand 14.5 → physically feasible 12.0 → commercial envelope 8.5. Each screen requires evidence.

03 | Examples are separate from sales

These market volumes illustrate the calculation. They are not the reported 2022 regional totals or actual Vopak deliveries.

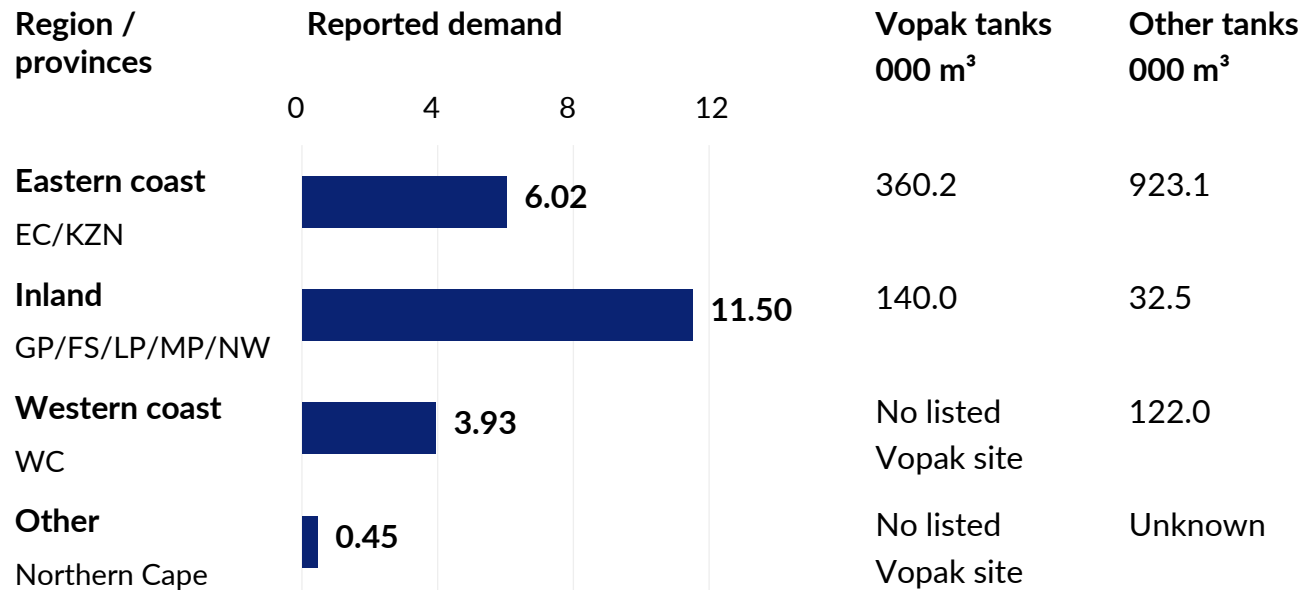
To resolve next

Nigel: request customer deliveries, destinations and transfer records. Manish: calculate unique served demand before estimating additional access.

Compare regional demand with Vopak's published storage footprint

All four regions | reported 2022 demand, bn litres/year

National total 21.90 bn L | petrol + diesel; jet excluded



Published capacity is a provisional footprint measure, not annual demand served.

Partial gross inventory; mixed products and unknown Sasol/Transnet capacities.

Fuel-compatible capacity share and actual Vopak demand share remain unestablished.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

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Regional competitor evidence

01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.

Transnet: Ladysmith lease tanks [7].

02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].

Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].
Transnet: Tarlton / Jameson Park + four lease sites [7].

03 Western coast

Burgan Cape: Cape Town petrol/diesel terminal [3].
Confirm current tank availability and customer access.

04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

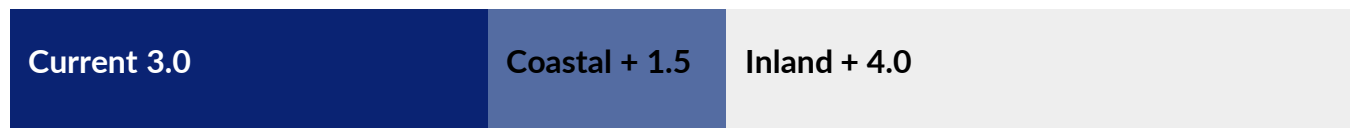
The illustrative additional opportunity is 5.5 billion litres a year

Accessibility gates and volume milestones | illustrative bn L/year

ILLUSTRATIVE | each increment requires evidence; not cumulative terminal receipts

5.5 bn L/year additional candidate

3.0 current example + 5.5 additional = 8.5 total envelope



What must hold before any additional litres are counted?

Competitive cost

Usable capacity

Customer access

Unique deliveries

Detailed customer-by-route gates belong in the evidence register.

Illustration only; 5.5 is neither a forecast nor a confirmed capturable market.

What the evidence shows

01 | Reach is necessary, not sufficient

Accessibility map p9 and competing routes p10 locate candidate access. Neither currently links a verified cost threshold to customer litres.

02 | Example milestones are conditional

The current example is 3.0. Coastal candidate 1.5 raises it to 4.5; inland candidate 4.0 raises it to 8.5. These are authored volumes.

03 | Every increment needs four gates

Evidence cost, route/tank capacity, customer rights and unique deliveries for the same product and period. Unassessed litres remain explicit.

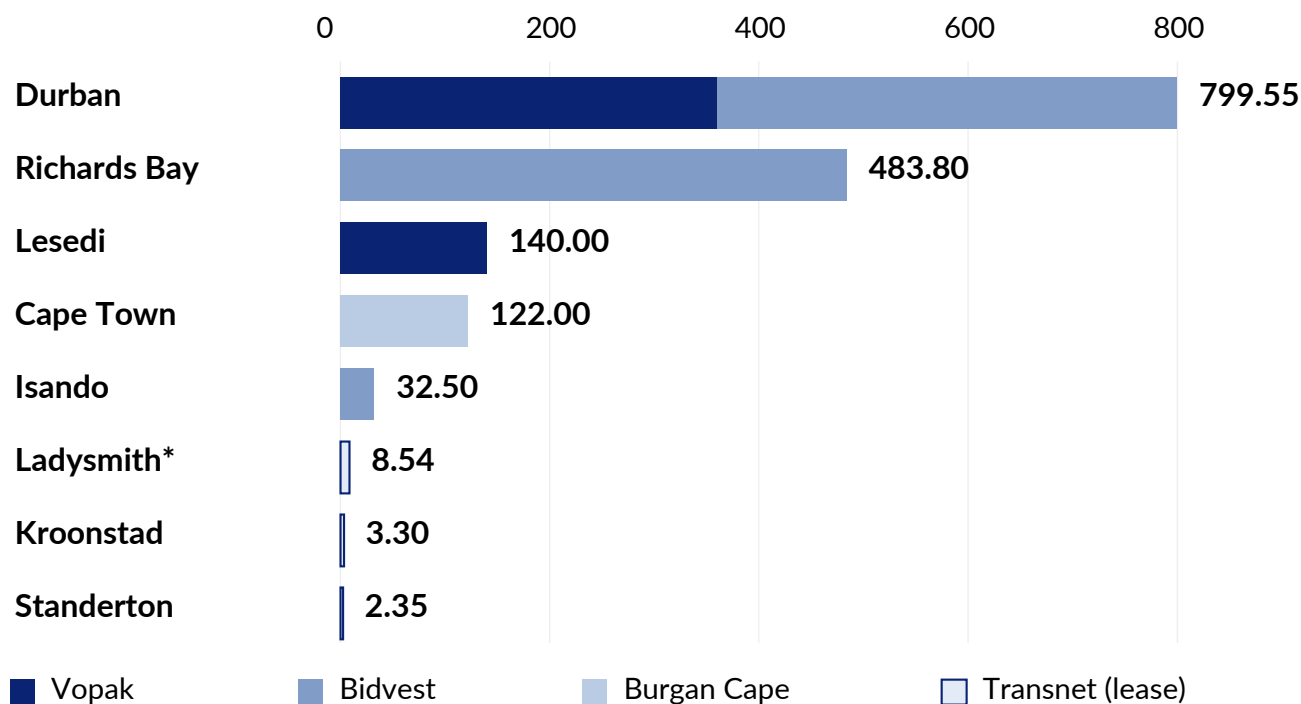
To resolve next

Nigel: agree destinations and acceptable delivered cost.
Manish: build a customer-by-route gate table; record passed, failed and unassessed litres, with sources and conditions.

Durban leads the published storage subset

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

*Ladysmith excludes 0.474 intermixture. Unquantified: Bethlehem, Magdala, Sasol, Tarlton, Jameson Park.

Source: Vopak; Bidvest; VTTI Burgan Cape; Transnet lease RFPs (5 Oct 2026), pp. 2-3. Checked 6 Oct 2026.

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What the evidence shows

01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

To resolve next

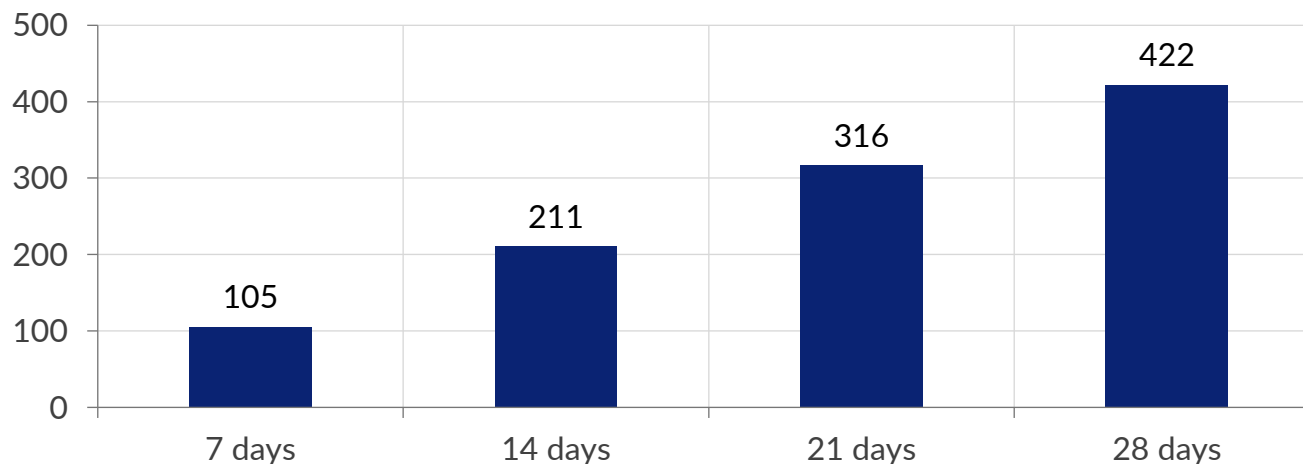
Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

Translate additional fuel flows into a preliminary inventory requirement

Working inventory sensitivity | illustrative, thousand m³

ILLUSTRATIVE | additional 5.5 bn litres/year; uniform daily flow

More inventory days require more working stock



Working m³ = annual litres ÷ 365 ÷ 1,000 × inventory days

At 14 days: approximately 211 thousand m³ of working inventory.

New tanks depend on usable spare capacity, peaks, product segregation and dispatch limits.

What the evidence shows

01 | Inventory days set working stock

At the illustrative 5.5 bn litres/year, 14 days needs about 211 thousand m³. More inventory days require more stock.

02 | Existing tanks may cover part

Usable spare tanks may absorb this inventory. Published gross capacity does not establish spare petrol/diesel space.

03 | Sizing needs an operating case

Test peak receipts, product segregation, tank heels and dispatch limits. This is a sensitivity, not an investment recommendation.

To resolve next

Nigel: confirm candidate flows and inventory policy.
Manish and Henry: test usable spare tanks, peak flows and operating allowances before sizing new capacity.

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com



Presentation palette | colours and HEX codes

Reference | colours used in this presentation

Core presentation colours

	Vopak blue #0A2373		Secondary blue #546CA2		Text / ink #000000		White #FFFFFF
	Unassessed / neutral #EEEEEE		Header divider #A0A0A0		Body divider #B8B8B8		Frame divider #B0B0B0

Sales-map colours | provincial totals, bn litres/year

	Below 1 #E3EBF6		1 to below 2 #BACCE4		2 to below 4 #809CC7		4 to below 6 #3D619D
	6 and above #0A2373		Map background #F5F8FB				

Other template theme colours

	Theme dark 2 #11151A		Theme light 2 #E7E6E6		Accent 3 #BDBEC1		Accent 4 #404040
	Accent 5 #767676		Accent 6 #D9D9D9				

Typeface: Lato. Sales colours show totals, not density or market share. Grey denotes unassessed demand.

Source: supplied Vopak template; pptx/brand_configs/vopak.py; provincial_demand_map.py.

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